

Learning an Interpretable Probabilistic Grammar of Hindustani Tāla: Mid-Semester Progress Report

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Abstract

We are building a probabilistic grammar of tabla rhythm in *tīntāl* whose derivation trees are themselves the explanation of each generated or analysed composition. This report covers the work completed so far on the 38-composition Mulgaonkar Tabla Solo corpus: a verified data pipeline, a timing-grounded representation, a supervised PCFG, an Inside–Outside EM study, an attribute layer that enforces cycle length and the *tihāī* landing constraint, a held-out study of parameter tying, and a generator that outputs a sequence, its derivation tree and its rule history together. Main findings: (i) the context-free grammar alone never produces a valid 16-mātrā cycle (0/2000 samples), and the attribute layer that fixes this also lowers held-out cost by 0.77 bits per stroke (about 15%); (ii) even EM started at the correct supervised probabilities drifts away from them, because parsing untimed bol strings is highly ambiguous; (iii) of 115 symbolic *tihāī* candidates, only 5 also satisfy the metrical constraint (at a 0.15-mātrā tolerance); (iv) *sam* is the only metrical position whose statistics measurably differ, and the *khālī vibhāg* shows no detectable effect; and (v) generated material repeats 4-stroke patterns far less than real compositions (median 3% vs. 68%). Baseline comparisons and human evaluation remain to be done.

1 Introduction

Hindustani rhythm is organised by *tāla*, a cycle (*āvartan*) of beats (*mātrā*) grouped into sections (*vibhāg*); *tīntāl* has 16 *mātrā* in four *vibhāg*s of four, with the cycle resolving on its first beat, *sam*, and its third *vibhāg* marked as *khālī* (Clayton, 2000). Tabla compositions are written and recited as sequences of stroke syllables (*bols*), and are built from named, hierarchical operations: theme and variation in a *kāyḍā*, and the *tihāī*, a phrase played three times so that it ends on *sam*.

This structure invites a grammatical treatment. Grammars of tabla already exist: the Bol Pro-

cessor encoded tabla idioms as formal grammars elicited from expert musicians (Kippen and Bel, 1989, 1992). Our aim differs in two ways. The production *probabilities* are learned from a corpus rather than elicited, and the grammar is compared quantitatively against black-box sequence models. The motivation is interpretability in the sense of Rudin (2019): a generated composition should come with its derivation tree, an explanation that is the model’s own production history, not a post-hoc account.

The project asks four questions. RQ1: does a corpus-fit grammar predict held-out compositions as well as matched-capacity black-box models? RQ2: do its high-probability rules correspond to recognised tabla units? RQ3: do derivation trees help people on rhythm tasks? RQ4: what formal power does *tāla* structure require? This report covers the groundwork for all four and the first answers to RQ4. Contributions so far:

- A verified, reproducible pipeline for the corpus, with a representation that places every stroke in the metrical grid using its onset time (§3).
- A supervised PCFG, a CYK/Inside parser and an Inside–Outside EM study showing that the untimed parsing problem is highly ambiguous (§4).
- An attribute layer (Knuth, 1968) that enforces cycle length and checks the *tihāī* arithmetic against real timing, and a constrained generator (§5).
- A held-out study of which metrical positions need their own parameters (§6), and an analysis of what the generator does and does not reproduce (§7).

2 Related Work

Grammars of tabla. The Bol Processor project modelled tabla improvisation with formal gram-

mers and used them in fieldwork with expert musicians (Kippen and Bel, 1992), and studied grammatical inference from expert-approved examples (Kippen and Bel, 1989). It is the direct precedent for this work; our delta is corpus-estimated probabilities, attribute constraints, and quantitative comparison with sequence models.

Probabilistic grammars of music. Generative syntax has been developed for tonal and jazz harmony (Rohrmeier, 2011), combined with a grammar of rhythm and metre, which improved the harmonic model (Harasim et al., 2019), and supported by a treebank of hierarchical analyses (Harasim et al., 2020). Minimum-description-length methods have been used to choose among musical structure models (Mavromatis, 2009). Probabilities of context-free grammars are classically estimated with the Inside–Outside algorithm (Lari and Young, 1990).

Computational analysis of Indian rhythm. Data-driven metre analysis of Indian art music is developed in Srinivasamurthy (2016). Our corpus comes from Gupta et al. (2015), who discovered syllabic percussion patterns in tabla solo recordings.

3 Data and Representation

Corpus. The Mulgaonkar Tabla Solo dataset (Gupta et al., 2015) has 38 *tintāl* compositions from six *gharānās* (Punjab 11, Benaras 9, Dilli 6, Ajrada 5, Lucknow 4, Farukhabad 3) and 20 composition types. Each comes with a score and two time-aligned onset files: one in a raw 40-symbol bol vocabulary, and one mapped by the dataset authors to 18 symbols. We verified that every score matches its raw onset sequence exactly, token by token (38/38). Compositions range from 47 to 611 onsets (mean 217.0). Rests (‘-’, 1,117 occurrences) appear only in the scores; the onset files have no rests and no durations. We use the 18-symbol vocabulary throughout, because many raw symbols are rare spelling variants of one stroke. The annotations are licensed CC BY-NC-ND 4.0; the audio is not redistributed.

Splits. Composition-level train/validation/test splits of 26/6/6 (seed 42). The two versions of one composition (*ajr_10_ajr*, *ajr_10_dli*) are kept in the same split to prevent leakage. The test split has not been used in any analysis reported here.

Comp.	<i>n</i>	Known tree	Inside	Δ
pjb_84	47	-184.45	-120.60	+63.86
ajr_12	96	-363.23	-252.52	+110.71
luc_17	164	-530.64	-348.96	+181.68

Table 1: Log-probability (nats) of the known timing-derived derivation vs. the Inside total over all parses of the same untimed bol string.

Representation. The hierarchy is Composition $\rightarrow$ *Āvartan* $\rightarrow$ *Vibhāg* $\rightarrow$ *Mātrā* $\rightarrow$ stroke. *Āvartan* boundaries come from score lines. The score notation does not reliably say which *mātrā* each token belongs to (group sizes vary with the subdivision), so instead each *āvartan* is divided into 16 equal *mātrās* and every stroke is placed by its onset time. This yields 351 *āvartans* and 8,245 strokes, with no onset lost. The equal-*mātrā* assumption is approximate: positions within a *mātrā* are imprecise (§5).

4 A Supervised PCFG and the Cost of Ambiguity

Grammar. Every non-terminal encodes a structural fact checked in earlier stages: Composition is a list of *Avartans*; an *Avartan* is four *Vibhag*_{*i*}; a *Vibhag* is four *mātrās*, the first of the cycle being *SamMatra*_{*i*}; and a *mātrā* is a list of *BolSlots*, each emitting one bol. Because each *āvartan*’s derivation is known from timing, probabilities are estimated by supervised maximum likelihood: 9 non-terminals, 44 rules (18 terminal). The named *kāyḍā* categories (*mukh*, *paṭṭā*, etc.) are not used yet: the corpus has no annotations from which they could be assigned reliably.

Parsing untimed strings is highly ambiguous. We implemented a CYK/Inside algorithm that sums over all parses. On three compositions its total log-probability far exceeds that of the single known derivation (Table 1). Without timing, the same bol string can be split into *mātrās* and *vibhāgs* in a very large number of grammatical ways. CYK runtime grew from 0.16 s ($n=47$) to 1.50 s ($n=96$) and 57.65 s ($n=256$), steeper than cubic. Real recordings therefore have to be scored with their timing-derived derivation, not by parsing.

Inside–Outside EM. The project plan prescribes EM, initialised from supervised estimates “to avoid poor local optima”. We tested this claim by running EM on all 351 *āvartans* from three initiali-

Initialisation	Iteration 1	Iteration 20
Supervised	-19037.83	- 19000.33
Uniform	-25597.38	-19009.39
Random (Dirichlet(1))	-35406.80	-19017.56

Table 2: Corpus log-likelihood (nats, 351 āvartans) under Inside–Outside EM.

sations (20 iterations, Dirichlet $\alpha=0.01$; Table 2). The predicted ranking held at every iteration, but the final gaps were small (about 0.1%). More importantly, *EM started at the supervised probabilities moved away from them*: for example, $P(\text{Avartan} \rightarrow \text{Vibhag}_2 \text{Vibhag}_3)$ went from 0.32 to 0.46. EM maximises the probability of the untimed string, which is dominated by spurious segmentations, so it is not a safe refinement here. We keep the supervised grammar.

5 Attribute Layer and the Tihāi Constraint

The grammar cannot count. Its list rules reuse a label ($\text{Avartan} \rightarrow \text{Vibhag}_0 \text{Avartan}$), so nothing tracks how many mātrās have been placed: 0 of 2,000 free samples were a valid cycle. Following attribute grammars (Knuth, 1968), every node carries a start position (inherited) and a duration (synthesised), in mātrās. Generation propagates the cycle constraint down the derivation: only learned productions consistent with “vibhāgs and mātrās still to place” are allowed, with their probabilities renormalised. This gives 2,000/2,000 valid cycles. For tīntāl the constraint fixes the whole skeleton; all remaining freedom lies inside the mātrās.

Empty mātrās. Only 117/351 real āvartans have a stroke in every mātrā, and 893 of 5,616 mātrās (15.9%) are empty. Our first grammar dropped them, losing the cycle length. Adding an explicit rest, using the score’s own ‘-’, makes all 351 real cycles well-formed. It also showed that the grammar’s “irregular” cycle rules came entirely from silent vibhāgs: with rests, *Avartan* has exactly three rules, each used 351 times. The rest symbol cannot tell silence from a sustained stroke.

Timing precision. Within a mātrā, only 17.5% of strokes fall within ± 0.05 of the beat (10% by chance), 10.8% near the half-mātrā (10% by chance), and quarter positions are below chance. Whole-mātrā placement is reliable; fractional positions are approximate. We therefore report real-data results at several tolerances.

Tol.	A	B	Regular	Strict (B+reg.)
0.10	8	17	53	4
0.15	8	18	65	5
0.25	9	19	87	12

Table 3: Real tihāi candidates ($N=115$) satisfying each condition at a given tolerance (mātrās). “Regular” means equal spacing between phrases. By chance, about 5 would land under B (4.3% of all onsets lie within 0.15 of sam).

The tihāi formula is ambiguous as written. The planned constraint $3d(p) + 2d(g) + 1 \equiv \text{target} \pmod{16}$ does not say what *target* is measured from or what the “+1” is. We tested two readings on real timing: (A) the last stroke of the third phrase is on sam; (B) the stroke after it is. Table 3 applies both to the 115 symbolic candidates (a phrase repeated three times with equal gaps). These came from an exact-repetition search over all 38 compositions.

Reading B lands at about $3.6\times$ chance, reading A at about $1.6\times$, so we adopt B. Only 5 candidates satisfy the full constraint at 0.15 mātrā. Three of those come from one long repeating passage (luc_18) that is probably not a single tihāi. The clearest match is in ben_25 (phrase ≈ 4.01 mātrā, gap ≈ 0.95 , next stroke 0.000 mātrā from sam; Figure 1). 55 candidates have no gap, which the plan’s allowed gaps $\{\frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1, 1\frac{1}{2}, 2\}$ exclude, and 25 of the 60 gapped ones fall outside that set. None of this has yet been checked by a tabla expert.

Generating tihāis. A tihāi cycle is generated by sampling a phrase of whole mātrās once, copying it three times, and placing it so the stroke after it falls exactly on the next sam. The three-fold copy is a dependency a context-free grammar cannot express for phrases of unbounded length. All 200 generated compositions pass the same checker used on real data. The symbol-only repetition search used to find real candidates recovers the exact boundaries in only 94/200. In 93 more it finds the same pattern shifted by one token, because a rest before the tihāi matches a rest gap on symbols alone; only durations separate the two. Without the attribute layer, 300 generated sequences contained no tihāi that landed on sam.

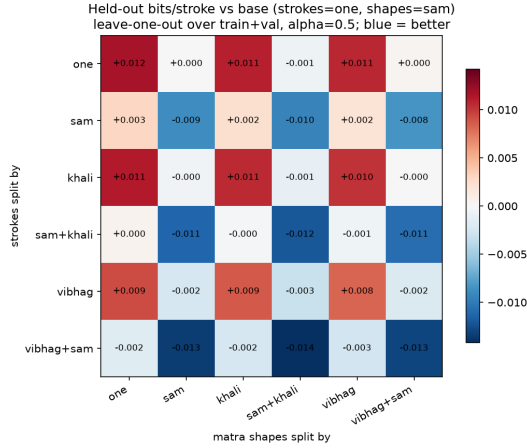


Figure 2: Held-out bits/stroke relative to the base for all 36 combinations of stroke tying (rows) and mātrā-shape tying (columns). Blue is better. Every improvement involves a sam split.

	Real	Generated
Empty mātrās	0.170	0.167
Mātrās with 1 stroke	0.391	0.397
Mātrās with 2 strokes	0.330	0.316
DHA share on sam	0.162	0.160
4-gram repetition (median)	0.680	0.034

Table 5: Generated vs. real (train and validation) statistics.

trained on this data, that only checks the sampler. It fails on repetition: in real compositions a 3-cycle window reuses a median 68% of its 4-stroke patterns, against 3% in generated ones. The grammar chooses each stroke from its position alone, so its only repetition is the tihāi copy.

8 Next Steps

- **Baselines and RQ1.** Train bol n -gram, LSTM and Transformer models with documented capacity, and compare them with the fixed grammar on the untouched test split. First, decide how the baselines are scored: the grammar’s cost covers strokes *and* their placement, whereas a bol sequence model predicts strokes only. Given §7, we expect the current grammar to predict worse than the baselines; this is a hypothesis to test, not a result.
- **Repetition.** Add productions that reuse and vary earlier phrases, which is what the kāyḍā vocabulary (mukh, dohrā, palṭā) describes, and measure the gain against the current grammar.

- **Validation.** Expert review of the tihāi matches and of generated output; a small seed treebank; and the planned behavioural study (RQ3).

9 Conclusion

A corpus-trained tabla grammar becomes usable only once duration constraints are enforced on top of it. Those constraints make every generated cycle valid, check tihāi arithmetic against real timing, and also improve held-out prediction. EM on untimed strings is unreliable because parsing them is highly ambiguous. Of the metrical positions, only sam measurably changes the statistics. The clearest gap is that real compositions repeat material and the grammar does not, which is the next modelling target.

Limitations

All results come from one corpus of 38 compositions by a single performer, in one tāla, using the dataset’s 18-symbol vocabulary; the 40-symbol vocabulary has not been evaluated. Metrical positions rely on an equal-mātrā assumption whose sub-mātrā precision is weak, so fractional results are reported at several tolerances. No musicological judgement here has been validated by a tabla expert, and the named kāyḍā categories are not yet modelled. The tihāi formula required an interpretive choice (reading B), made on corpus evidence. No baseline comparison or human evaluation has been run yet; claims about RQ1–RQ3 are therefore deferred. The implementation is checked by 65 regression tests, several of which caught real errors during development before any results were reported.

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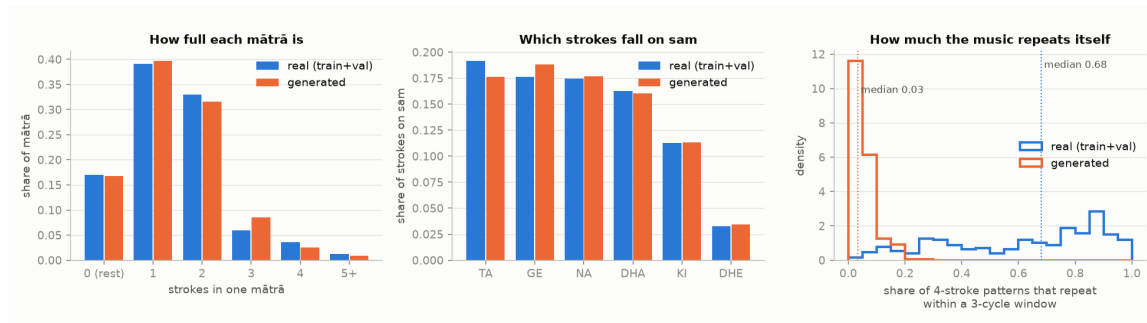


Figure 3: Generated (orange) vs. real (blue). Left and middle: quantities the grammar models. Right: share of 4-stroke patterns that recur within a 3-cycle window. Real tabla is strongly self-repetitive; the generator is not.

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